

Project Documentation Notice: This document is created specifically for the Week 7 IoT, Connectivity & RTOS portfolio project. It details the operational theory, single-bus digital signaling protocol, calibration data, and ESP32 interface implemented in Tasks 4 and 5.

1. Sensor Specifications & Operating Envelope

Parameter	Measurement Range	Accuracy	Resolution	Sampling Period
Temperature	0°C to 50°C (32°F to 122°F)	±2.0°C	1°C (8-bit)	1 second (1 Hz max)
Relative Humidity	20% to 90% RH	±5.0% RH	1% RH (8-bit)	1 second (1 Hz max)
Operating Voltage	3.0V to 5.5V DC	Recommended: 3.3V	—	Standby: 100uA

2. Single-Bus Communication Protocol

The DHT11 utilizes a proprietary single-bus bidirectional protocol on a single digital pin (GPIO 4 on the ESP32), pulled high via a 10kΩ resistor:

- Host Start Signal:** ESP32 pulls the data line low for at least 18ms, then high for 20–40μs.
- Sensor Response:** DHT11 pulls low for 80μs followed by 80μs high, acknowledging presence.
- Data Transmission (40 Bits):** Byte 1: Humidity Integer, Byte 2: Humidity Decimal (0), Byte 3: Temperature Integer, Byte 4: Temperature Decimal (0), Byte 5: Checksum = (Byte 1 + Byte 2 + Byte 3 + Byte 4) & 0xFF.
- Bit Encoding:** Every bit starts with a 50μs low pulse. Logic 0 is followed by 26–28μs high; Logic 1 is followed by 70μs high.

3. ESP32 Driver Integration & Reliability Rules

Due to the microsecond timing constraints of the single-wire protocol, interrupts are briefly suspended during bit decoding. A hardware watchdog interval of 2000ms ensures compliance with the 1Hz minimum sampling interval, preventing thermal self-heating and inaccurate readings.